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Combating Money Laundering using Artificial Intelligence

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Abstract

This research provides a comprehensive outline of money laundering, its cycle, and the challenges of detecting it, in Nigeria and globally. It argues that traditional, rule-based models for identifying financial crimes are inefficient as a result of their high false-positive rates and static nature. The research proposes a solution leveraging modern machine learning and deep learning, specifically an unsupervised approach using a clustering model. This methodology aims to identify suspicious transactions during the "placement" stage of money laundering by detecting anomalies and evolving patterns. Developing and assessing a generative deep learning model for fraud detection, assessing the likelihood of financial crimes, and contrasting the suggested methodology with conventional methods are the goals of this research. The paper's objectives are to create and evaluate a generative deep learning model for fraud detection, analyse the risks of financial crimes, and compare the proposed method against traditional approaches.

Keywords: *Clustering; Deep Learning; Machine Learning; Money Laundering; Terrorism Financing; Artificial Intelligence.*

1. Introduction

Money Laundering entails camouflaging illegally obtained funds as legitimate by combining them into the financial system. Predicate crimes, which vary by country, form its basis. (Odufisan et al., 2025) [1] investigated the possibilities of Artificial Intelligence (AI) and Machine Learning (ML) for intensified fraud detection and avoidance in Nigeria. They highlighted the interests of AI-powered fraud detection, as well as enlarged efficiency, better accuracy, and proactive risk lightning. According to Lessambo (2023) [2], global approaches to anti-money laundering and counter-financing of terrorism in the banking sector show the restrictions of static, rule-based systems and expose the demand for more adaptive strategies.

In conformity with the Financial Action Task Force (FATF), money laundering entails concealing the illegal origin of funds obtained through a predicate crime, an offense that generates illicit proceeds. The specific crimes that qualify as predicate offenses vary from country to country. Money Laundering in Nigeria is defined as the act of committing a predicate crime, obtaining illegal funds, and then attempting to hide the source of those funds.

Predicate crimes in Nigeria include human trafficking, terrorism financing, sexual exploitation, drug and arms trafficking, corruption, bribery, fraud, tax crimes, extortion, smuggling, environmental offenses, murder, piracy, and insider trading. Financial and non-financial organizations must record suspicious transactions for investigation to prevent illicit funds from entering the financial system. Rule-based systems, while straightforward, produce many false positives and are costly to manage, as they are easily exploited by bad actors. Machine learning and deep learning offer improved accuracy by identifying suspicious patterns and reducing false positives, making them more efficient than traditional rule-based methods.

1.1. The cycle of money laundering

Positioning, grouping, and unification make up the money laundering cycle as shown in **Figure 1**. Placement involves depositing illicit funds into financial institutions while escaping detection by anti-money laundering (AML) regulations. Criminals use strategies like structuring (or smurfing) to circumvent financial scrutiny and bypass AML / Combating the Financing of terrorism (CFT) measures [2].

Layering is a convoluted web of transaction movement in the financial system that usually involves using mules to move money from one account to another to confuse the authorities from suspecting that the funds being moved are proceeds of fraud and criminality. Funds can be moved between accounts, people, and

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different nations.

Paper transactions, electronic transactions like wire transfers, and/or the manual transfer of funds between nations through secret means can all be considered forms of layering. Non-traditional financial systems may also be used to launder the money. That is, funds may flow more quickly than the authorities can track and may take the shape of different investments.

The last phase of money laundering is integration, where illicit funds, after being placed and layered within the financial system, are returned to the criminal from seemingly legitimate sources. This can involve the criminal profiting from businesses acquired with the laundered money, such as restaurants, stores, or repair shops, effectively disguising the money's illegal origin. To avoid detection, the bad actors may use the money to start businesses by closely adhering to all industry and governmental regulations, such as timely filing of tax returns and careful payment of all business and employee taxes.

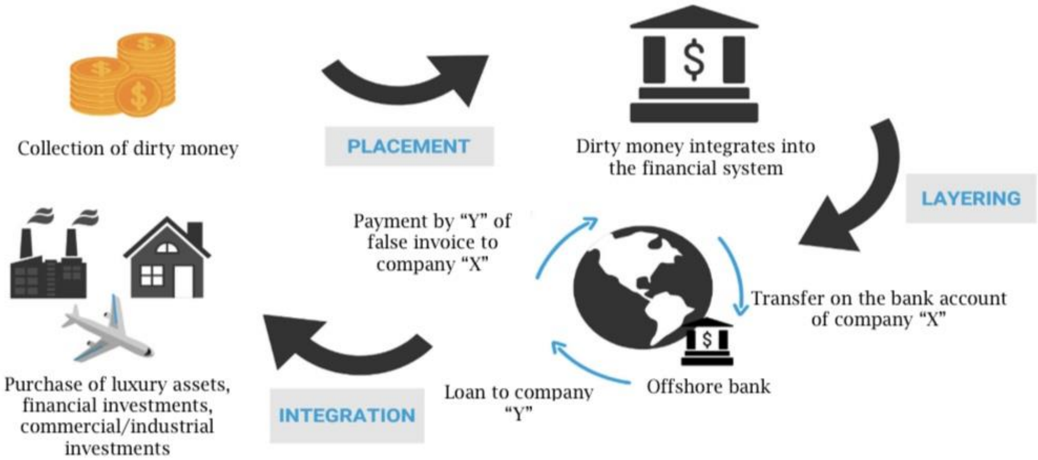

Figure 1. *Money Laundering Cycle*

1.2. The definition of the problem

Money Laundering and terrorism financing undermine a nation's economic stability, often linked to crimes like goods dumping that harm local economies. These activities also risk banks and non-banking institutions losing their correspondent banking relationships, which can impact their operations and financial stability. The activities of money laundering and terrorism financing also affect communities that are endowed with natural resources. Sometimes, these bad actors are responsible for the illegal activities that occur in these communities, ensuring that the communities' resources are used for their own selfish ends. They also cause chaos by inciting crises among the community members, which detracts from their activities.

Governments and international organizations, including Nigeria and globally, are collaborating to combat money laundering and terrorism financing. Traditional rule-based systems are ineffective due to frequent changes and high false positives, making monitoring costly and inefficient. Artificial intelligence offers a solution by improving accuracy, reducing false positives, and lowering functional costs for financial organizations, letting them detect suspicious trends and support investigations.

The decline in correspondent banking relationships (CBRs) between emerging and developed markets is driven by concerns over AML/CFT compliance. Correspondent banks are increasingly ending ties due to challenges in assessing risks related to customer due diligence (CDD) and Know Your Customer (KYC) programs. As CBRs are vital for global financial connectivity, respondent banks are investing in robust KYC and CDD measures to address these concerns and combat money laundering and terrorism financing effectively. Financial institution faces the following risk categories due to ineffective AML/CFT strategies (**Figure 2**):

- **Business/Strategic risk:** Business strategy and strategic objectives risk created by an organization's operations. These risks are associated with the organization's products and services.
- **Market risk:** Unfavorable changes in market prices or rates, or foreign exchange rates, affect a bank's financial health.
- **Credit risk:** The risk associated with counterparties or borrowers with poor credit who will not fulfill their obligations or perform on them.

- **Liquidity risk:** When a business can't pay its debts because it doesn't have enough money, can't sell off assets or get enough money, or can't unwind or offset a particular exposure without materially altering its capital or balance sheet levels. Liquidity risk affects the organization.
- **Operational risk:** This risk is linked to unenforceable agreements, court cases, or unfavorable rulings that could interfere with or negatively impact a banking organization's operations or state.
- **Reputational risk:** Whether or not it is true, bad press about an organization's business practices can result in revenue losses, expensive legal action, or a drop in the institution's clientele.
- **Compliance risk:** Risk associated with non-adherence to industry and government laws and regulations, internal policies, or recommended optimal methods. Banks run the risk of incurring material losses, financial forfeitures, and legal penalties.

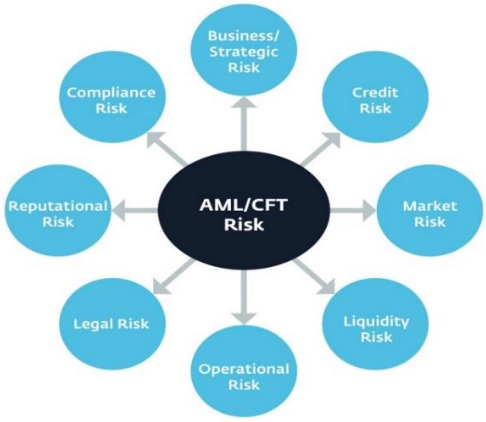

Figure 2. *AML/CFT Risk Relationship Chart.*

The compliance or legal risk classification is where AML/CFT risks are mainly included. Liquidity, strategy, operations, legal/compliance, reputation, and, in certain cases, credit risk are among the risk categories that can be impacted by AML/CFT risks.

1.3. Purpose

Rule-based systems for detecting financial crimes produce up to 98% false positives, making investigations time-consuming and inefficient. These systems are also vulnerable to exploitation by bad actors. Cutting-edge technology, like natural language processing (NLP), deep learning (DL), and machine learning (ML), offers improved detection by identifying patterns, anomalies, and risks with minimal human intervention. ML enables systems to gain understanding from data, reduce false positives, and enhance risk management. DL, a more advanced ML form, refines detection capabilities further. This research seeks to leverage generative deep learning and clustering techniques to combat money laundering and financing of terrorism during the placement phase.

1.4. Aim and Objectives

This research seeks to design an anti-money laundering model that can identify money laundering activity while it is still in the placement stage. The following are the study's particular goals:

- A thorough literature review to understand the present status of study in the struggle averse to money laundering and the interest in artificial intelligence, particularly machine learning and deep learning, in preventing money laundering.
- Determine the kind of risk that financial institutions are exposed to due to money laundering and terrorism financing operations.
- Analyze how the political, social, and economic life of a country like Nigeria and globally is affected by the actions of money launderers and those who finance terrorism.
- Analyze and identify traditional fraud detection methods' flaws and deficiencies.
- To get the deep learning models ready. Using deep learning, create a generative model for anti-money laundering detection.
- Use the latest datasets to test and assess the suggested model

2. Related Works

[3] focus on risk management associated with the bank’s clients and products by concentrating on transaction monitoring in the fight opposed to money laundering. Although artificial intelligence and machine learning were proposed to automate the transaction monitoring process, regulators and financial institutions prefer a straightforward system that is simple to understand for stakeholders. The authors argue for explainable artificial intelligence (AI), which provides stakeholders with concise, understandable results so they can make decisions quickly.

Similarly, Tang et al. (2025) [4] proposed an algorithm for identifying suspicious conducts in big payment circulations derived from deep generative frameworks. By joining Generative Adversarial Networks (GAN) and Variational Autoencoders (VAE), the algorithm is created to discover unusual conducts in financial transactions. The study gives a detailed analogy of performance in identifying several transaction structures (like money laundering and fraud) in big payment circulations, authenticating the precedence of generative models in managing large financial datasets. [5] focused on the unifying of terminology in two areas of the Anti-money laundering fight, namely:

- Risk assessment of clients
- Flagging of suspicious activity

Client risk profiling examines risk factors that are associated with customers of a financial institution, and suspicious behavior flagging examines the red flags that will help banks to find illicit transactions among the millions of legal transactions.

A comprehensive view of the clients was introduced in 2021 as part of an attempt to deploy a strong AML artificial intelligence solution. This approach aims to effectively know the clients for anti-money laundering purposes. To generate a truly unified view of a client's risk profile, banks are advised to combine their frequently fragmented client data across various teams, business processes, and systems, including KYC, CDD, EDD, sanction screening list, negative news screening, transaction monitoring, and others. The actual business and compliance needs should be determined by the client's consolidated view. A comprehensive and dynamic assessment of a client using all available data, both internal and external. Merging transaction data, funding sources and uses, sanction lists, multilingual news articles, and CRM data.

Delivering almost real-time updates to the data image. Preserving the financial flows, associations, and connected parties in the network graph. By adding more intelligence layers through sophisticated rules or AI/ML, the solution greatly lowers risk and offers enormous business operational benefits. Stakeholders and industry regulators greatly value the transparency of the artificial intelligence output produced by this method.

A total of 9,682 (nine thousand, six hundred and eighty-two) SARs were received from the different sectors of Nigeria's financial industry, with the stockbroking firms receiving the fewest, according to [6] (**Table 1**). Fewer than ten SARs were reported by BDCs, merchant banks, and other financial institutions in 2019.

Table 1. *(NFIU 2019 report) NFIU sectoral report of SARs from the various sectors.*

S/N	Agency/Business Type	Total Number of Sars Reported
1	Banks	9,525
2	Insurance Companies	120
3	Virtual Asset Providers	21
4	Merchant Banks	6
5	Other FIs	5
6	Bureau De Change	4
7	Stockbroking Firms	1
	Total	9,682

In 2019, a total of 12,716 (twelve thousand, seven hundred and sixteen) STRs totaling N576,103,016,428 were received by NFIU. STRs were most prevalent in the banking industry, with 12,457 (twelve thousand, four hundred and fifty-seven) reported. The insurance industry submitted 61 reports from merchant banks and 147 reports from insurance companies.

Table 2. *(NFIU 2019 report) NFIU sectoral report of SARs from the various sectors by volume.*

S/N	Sector	Total Number of Strs Received	Total Volume of Transactions (N)
1	Banks	12,457	557,193,536,032
2	Insurance Companies	147	1,830,619,993
3	Merchant Banks	61	13,164,517,792
4	Primary Mortgage Banks	41	171,815,358
5	Stockbroking Firms	3	3,604,800,000
6	Issuing Houses	3	51,500,000
7	Other Financial Institutions	2	52,569,699
8	Insurance Broker	2	33,657,554
	Total	12,716	576,103,016,428

The NFIU report in **Table 2** focuses on the operational activities of the organization, including the quantity of reports it receives and examines. While the report acknowledges the importance of conducting enhanced due diligence on Politically Exposed Persons (PEPs), it does not provide details on specific cases involving them. Instead, it emphasizes the exploration of Suspicious Transaction Reports (STRs) and other financial expertise.

The Economic and Financial Crimes Commission (EFCC) reports that 41 people were convicted of money laundering in 2022, a sharp rise from 9 in 2021 and 5 in 2020 [7], [8], [9]. A report from the Nigeria Sanction Committee identified 13 people and six organizations as terrorist financiers [10].

The research proposes an unsupervised anomaly detection approach to combat money laundering and terrorism financing. It uses a clustering deep learning model to control the constraints of traditional, rule-based models and supervised machine learning. By not relying on pre-labelled data, this method is better suited to detect the hidden and unfolding structures of financial crime.

3. Proposed Method

3.1. Dataset Preparation

The primary goal is to identify suspicious transactions without prior labels. The process involves:

Data Preparation: The SAML-D.csv dataset (over 1 million data points) is preprocessed by handling missing values, converting categorical data, and using SMOTE to handle class imbalance.

Anomaly Detection: A clustering model is trained on “normal” transaction data. The system then uses the model’s prediction errors to find transactions that don’t fit the expected pattern. An anomaly threshold (e.g., the 95th percentile of errors) is set to flag suspicious activity. [11], [12].

Data preparation steps involve data cleaning, data harmonization, feature selection, and data transformation [12]. Considering that the financial data is sensitive, financial organizations are not willing to share their data with researchers. As a result, researchers use synthetic data to obtain the data they need for their research. A computer program creates synthetic data instead of gathering it from actual sources. There are several ways to enhance machine learning models while preserving actual data privacy using synthetic data [12], [13].

Synthetic data can be utilized to augment or substitute traditional data providers in the financial services industry, giving a more comprehensive picture of the financial institution’s underlying risk to a suspicious customer’s actions. For this research, synthetic data was used to create the data for the research. Three tables, customer, account, and transaction tables respectively, were created and later merged these tables into one table for the training and testing of the model.

3.2. Methodology

The research’s design methodology includes creating a deep learning approach for identifying wary deals with money laundering, creating a clustering technique to describe the choices built by deep learning approaches, and creating a traditional machine learning method to identify wary transactions for the performance of the deep learning approach can be in comparison to that of traditional machine learning approaches.

Banks, Insurance companies, FinTechs, Casinos, and other reporting entities are responsible for identifying suspicious money laundering deals and reporting skeptical transactions to the Nigeria Financial Intelligence Unit (NFIU). The NFIU will then look into the suspicious transactions reports (STRs) and suspicious activities reports (SARs). If the NFIU officers are convinced of suspicious money laundering and terrorism financing, Further investigation will be carried out, and the cases will be reported to the law enforcement Agencies (LEAs) (i.e. The Nigerian Police, EFCC, and ICPC) for further investigations and possible prosecution in the suitable court of law.

Financial institutions provided the following data, which were used to spot any suspicious money laundering activity and alert the appropriate authorities to guarantee adherence:

- ***Client:*** Bank customer data, including risk assessment, BVN, NIN, KYC, CDD, and EDD. The National Identification Management Commission is the government agency responsible for issuing the National Identification Number (NIN). The Central Bank of Nigeria has a middle location for the Bank Verification Number (BVN).
- ***Products:*** The things that the customer owns, such as savings, credit, and loan accounts.
- ***Transactions:*** The financial exchanges involving the client's possessions.
- ***Transaction logs:*** These examine how logs from banking systems are used to track transactions and detect anomalous flows. Requirements such as explainability, flexibility and identifying new risks in transactions were extracted and analyzed (Öztaş, 2024) [14].

Every position a Politically Exposed Person (PEP) holds is categorized by risk level to assist the bank in evaluating the degree of risk. The risk level for each type of position, which is classified using the following system, is listed below:

High risk: Politically Exposed Persons (PEPs) in Class 1

- Heads of State and Government.
- Members of National and Regional Governments.
- Members of National and Regional Parliaments.
- Heads of the armed forces, law enforcement, and legal systems.
- Central Bank board members.
- Political party senior representatives.

Moderate risk: (PEPs) in Class 2

- Top officials from the military, judiciary, and law enforcement.
- Top civil servants, other state officials, and agencies.
- Participants of senior religious communities.
- Consuls, high commissioners, and ambassadors.

Medium risk: PEPs in Class 3

- The top management and board of directors of state-owned companies and establishments.

Low risk: PEPs in Class 4

- Mayors and representatives in local, state, municipal, and district gatherings.
- Top managers and employees of multinational or supernational organizations.

To identify and manage financial and regulatory risks, numerous organizations worldwide rely on the Refinitiv Company's World-Check data, a directory of PEPs, high-risk persons, and enterprises.

Social media data contains data from various social media sites, like X (formerly Twitter), Facebook, LinkedIn, and others. This research's objective is to build a deep learning framework capable of successfully spotting questionable money laundering transactions. To identify wary money laundering deals, classifications of machine learning like Decision Tree, SVM, Naïve Bayes, RF, Logistic Regression, XGBoost, and Clustering are commonly used, according to the literature review. To compare the results of clustering machine learning classifiers based on prediction accuracy and frequency of use, three classifications are derived from the list. These classifiers consist of RF, SVM, and XGBoost.

3.3. Clustering

An unsupervised learning method called clustering is used to organize data points into clusters following their likeness or distance in feature space. Data points in each cluster are more alike than they are with points in other clusters.

Clustering algorithms include density-based, centroid-based, and distribution-based methods, utilizing distance measures like Manhattan and Euclidean distances to calculate similarity scores. Clustering can be categorized as hard clustering, where objects belong distinctly to one class, or soft clustering (fuzzy clustering), which assigns probabilities to an object belonging to specific clusters [15].

K-Means clustering assigns observations to the closest cluster center, reevaluates centers as the average of assigned points, and iterates up to the time when data points no longer shift clusters. It aims to minimize within-cluster variation and maximize differences between clusters, producing homogeneous clusters internally while ensuring heterogeneity among clusters.

The data points' variability to their distance from the cluster centroid is measured by WCSS (With the cluster sum of squares or Cohesion). WCSS determines the separation between the centroid of each cluster and a data point. Every data point within that cluster goes through this process again. The sum of the results is then segmented by the cluster's total number of points. The procedure is replicated for every cluster. A final step involves averaging the outcomes for each cluster. While a lesser value suggests that the data points are fairly comparable and uniform and that the cluster is tight, a larger value indicates a wide data spread. Another name for the intra-cluster distance is the cluster's inertia. The total of all the distances is what it is. The better the cluster, the lower the inertia value.

Inter-cluster sum of squares: This metric calculates the separation between each cluster's centroids. The average value is obtained by dividing the distance between each cluster's centroids by the total number of clusters. The better the clustering, the larger it is, suggesting that the clusters are diverse and easily distinguished from one another.

Silhouette Value: The silhouette value is used to assess the effectiveness of clustering. A higher value is preferable, and it ranges from -1 to +1. For every observation, we compute the standard distance from every data point within the same cluster, say $\boldsymbol{x\iota}$, to determine the degree to which an observation resembles the items in its cluster with other clusters. Next, we determine the average distance, let's call it $\boldsymbol{y\iota}$, between each data point in the nearest cluster. Next, we employ the subsequent equation below to derive the coefficient.

$$Silhouette\ Coefficient = (y\iota - x\iota)\ /\ \max(y\iota, x\iota) \quad (1)$$

A value of -1 for the coefficient indicates that the inspection is in the incorrect cluster. If it is zero, the inspection is quite near the nearby clusters. When the coefficient is +1, it indicates that the inspection is far from the nearby clusters.

Therefore, we shall anticipate a good clustering solution if the coefficient had the largest value.

Dunn Index: The effectiveness of the clustering is checked making use of the Dunn Index. Equation 2 below provides the KPI, defined in the inter-cluster sum of squares and Silhouette Value.

$$Dunn\ Index = \min(\text{inter} - \text{cluster distance}) / \max\ (intra - cluster\ distance) \quad (2)$$

It is best to maximize the Dunn index value. To maximize it, the lowest possible denominator should be used, indicating that the clusters are fairly strong and closely filled, while the numerator should be as large as possible, suggesting that the clusters are far apart.

Determining the best value of "k": The Elbow method is the extensively used approach for getting the ideal value of "k" in the K-Means clustering method. For many values of "k", we compute using the method in the cluster sum of squares, or WCSS. A graph is then created by plotting WCSS against various "k" values. Whenever an elbow or kink is seen. The dataset's ideal number of clusters is this one.

Hierarchical clustering is another name for connectivity-based clustering. The root is the initial node and is subsequently divided into nodes and sub-nodes iteratively. A node is referred to as a terminal node or leaf if it cannot be further divided. The Equation below allows us to create many dendrograms because there are multiple processes or logics to combine the observations into clusters.

$$Number\ of\ dendrograms = (2n-3)!\ /\ [2(n-2)(n-2)!] \quad (3)$$

To identify the perfect number of clusters:

K-Means Clustering: The elbow method gets the ideal number of clusters by observing where the within-cluster variation declines at a far slower rate.

Hierarchical Clustering: Dendrograms help determine the number of clusters. The altitude at which two clusters merge represents their distance, and the clusters are indicated by the number of vertical lines that a horizontal line touches.

Density-based clustering: Clusters are zones of heavy points parted by minor areas, with outliers or noise in the sparse regions.

An activation function known as the Rectified Linear Unit (ReLU), defines an argument's positives. The ReLU activation function is displayed below. Remember that even for negative values, the value is zero, and it begins to rise from zero.

$$F(x) = \max\ (0, x) \quad (4)$$

For example, if x is positive, the output will be 0

One of the activation functions employed in neural networks is the ReLU function, which is displayed in it. ReLU is less costly to train and easier to use (**Figure 3**).

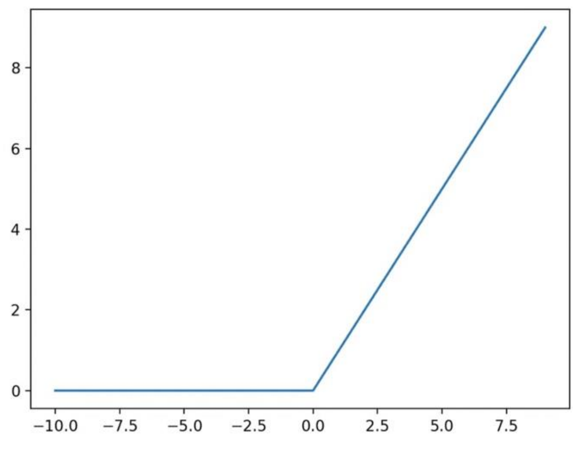

Figure 3. *ReLU function.*

Due to its simplicity, it can be computed much more quickly and at a lower cost. It is not centered at zero and has no boundaries. Except for zero, it is differentiable everywhere. ReLU is frequently used in hidden layers to train networks more quickly because it is a less complex and computationally expensive function.

This is the fundamental framework for a network designed to describe the procedure. Data transformation layers (i.e. Hidden layers) receive information from the input layer. The activation functions, weights, and bias terms were used to process the data in the hidden layers. After that, a guess is made using the data set. Because the input variables are evaluated sequentially between the input and output layers, this process is known as feed-forward propagation. The algorithm is fed this training dataset. After that, the network will learn the different attributes of input data and comprehend the attributes of inputs. The algorithm can then predict the inputs based on this training. If the input is part of a group or cluster, the prediction will be a probability score. We can declare with confidence that the input is a member of the cluster or group if the probability is high enough.

Once the hidden layers' data processing is finished, a forecast, or the likelihood that the input belongs to a particular group or cluster, is produced. The loss function, sometimes referred to as the objective function, can be used to quantify accuracy in the network. The predicted and actual values are contrasted by the loss function. By calculating the difference score, the loss function may determine the error rates and the network's performance.

The goal of our solution is to maximize accuracy by minimizing this inaccuracy. After that, the loss score (Predictions – Actual) is utilized as input to modify the mass's values to continuously lessen the inaccuracy. The backpropagation algorithm is used to accomplish this task.

Backpropagation is utilized to adjust the connection loads, whereas the rate of improvement determines the magnitude of the corrective actions to lower the inaccuracy. Based on the error, these weights are updated backward; after that, the corresponding weights are modified, gradient descent is computed, and the errors are recalculated. For this reason, back-propagation is occasionally referred to as the core algorithm for deep learning. The backpropagation process, in which data moves between the output and hidden layers, is depicted in the image below. It should be noted that this flow is backward in contrast to further propagation, which moves data from left to right. The weights are initially assigned some random values to begin the training process. An initial output is produced using these arbitrary values. Since this is the first attempt, the output may differ significantly from the actual values, and as a result, the loss score is extremely high. Weights (and biases) are slightly shifted in the right direction during neural network training, which lowers the loss score. The most ideal weight values that minimize the loss function are obtained by repeatedly iterating this loop. During the network training process, back-propagation enables us to decrease the error iteratively.

A loss function is computed to optimize a neural network. The loss function indicates the level of discrepancy in the midst of the predicted values and the actual values. Backpropagation determines the gradient between each weight and the loss function. The loss can be gradually decreased by updating each weight separately over iterations using this information. Back-propagation involves calculating the gradient backwards that is, from the network's last layer through its shrouded layers to its first layer. To get the gradient of any layer, the gradients of each layer are added together making use of the calculus chain rule. We will now discuss the procedure in greater detail. First, let's indicate some mathematical symbols:

1. hi) - the hidden layer I's output
2. gi) - the hidden layer I's activation function
3. wi) - the layer I hidden weights matrix
4. bi) - the layer I bias
5. $\boldsymbol{x}$ - the input vector
6. N - the network's total number of layers
7. The network weight from node j in layer (i-1) to node k in layer I is wi)jk
8. ∂A/∂B - it is the partial derivative of A to B

The input $\boldsymbol{x}$ is supplied into the network during training, and it moves between the layers to produce the output Þ. y is the anticipated result. Therefore, C (y, η) will be the loss or cost function to compare y and η. Additionally, any network's hidden layer's output can be shown as follows:

$$h(i) = g(i)\ (w(i)T\boldsymbol{x} + b(i)) \tag{5}$$

where i (index) can represent any network layer. The final layer's output is:

$$y(x) = \mathrm{w(N)T\ h(N-1)\ +\ b(N)} \tag{6}$$

We change the network's weights during training to lower C. We determine each network weight's derivative of C. For each network weight, the derivative of C is as follows. We are now aware that a neural network consists of numerous layers. The back-propagation algorithm begins by computing the derivatives at the Nth layer, the network's final layer. After that, these derivatives are a little backwards. The (N-1) layer of the network will thus receive the derivatives at the Nth layer, and so forth.

The calculus chain rule is used to determine each component of the derivatives of C separately. Therefore, the derivatives of layer N are utilized in layer (N-1) in the back-propagation process to save and reuse them in layer (N-2). Using the derivatives of the most recent computations, we obtain the derivatives of the current layers by starting with the network's last layer and working our way through all the layers to the first layer. As a result, back-propagation proves to be far more effective than a traditional method in which we would have determined each network weight separately. We will update all of the network's weights after determining the gradients. The goal is to reduce the cost function.

Feeding autoencoders is the input is squashed into a lower-dimensional code by autoencoders, which are feed-forward neural networks, and then they try to recreate the output from this representation. Learning the lower-dimensional representation (i.e., encoding) for a high-dimensional dataset is the aim of an autoencoder. Keep in mind the Principal Component Analysis (PCA) discussed in the earlier chapters. One could consider autoencoders to be a PCA generalization. Autoencoders are capable of learning nonlinear relationships, while PCA is a linear approach. Autoencoders are therefore necessary for dimensionality reduction solutions, which extract the most important characteristics from the input data. An autoencoder's architecture is very easy to comprehend. Three components make up an autoencoder: a decoder, a bottleneck or code, and an encoder. To put it simply, an encoder compresses the input data, the decoder decompresses the knowledge and restores it to its original form, and a bottleneck or code contains the compressed information. The input and output can be compared after the data has been decompressed and reconstructed to its encoded form.

Encoder: The encoder processes the input data. A fully connected artificial neural network is all that an encoder is. The output is produced as a result of compressing the input data into an encoded representation.

Bottleneck: The encoder's bottleneck is sometimes referred to as its brain. It has compressed information representations, and the bottleneck's job is to let only the most crucial information through.

Decoder: A decoder breaks down the information that was received from the bottleneck. The data is restored to its original or encoded state. After a decoder completes its task, the decompressed values it produces are compared to the actual values.

A few crucial considerations for autoencoders are as follows:

- When decompressing autoencoders, information is lost compared to the original inputs. Therefore, there is a loss when the compressed data is decompressed in comparison to the original data.
- Autoencoders are dataset-specific. This implies that an algorithm trained on traffic signal images won't work on flower images, and vice versa. This is because the autoencoder would have only learned features unique to flowers. Thus, we can conclude that auto-encoders can only compress data similar to that used for training.
- Specialized instances of algorithms can be trained to perform well on particular input types comparatively more easily. All that is required to train the autoencoder is a representative training dataset. Now we have discussed the main parts of autoencoders.

The procedure is as follows:

- Initially, the encoder module interprets the input data.
- The input to a model is compressed into a small bottleneck by the encoder. Because it limits information flow and permits only critical information to flow through, the bottleneck is sometimes referred to as knowledge representation.
- The decoder, which comes after the bottleneck, decompresses the data and restores it to its original or encoded state.

To extract the most important characteristics from the input data, this encoder-decoder architecture is highly effective. Producing an output that is the same as the input is the goal of the solution. The decoder architecture is typically seen as a mirror image of the coder architecture. Although not required, it is typically adhered to. We guarantee that the inputs and outputs have the same dimensionality. Code size is possibly the most important hyperparameter. The count of nodes in the middle layer is represented by it. The data compression is determined by this, which can also serve as a regularization term. The data is more compressed, the smaller the code size value. The autoencoder's depth is indicated by a parameter called the number of layers. A more detailed model is more complicated and will require more processing time. They only require neural networks as a component of their overall framework. The weight used per layer is the number of nodes. As the input gets smaller across the layers, it usually gets smaller with each new layer. Additionally, it rises in the decoder.

The loss function is the last hyperparameter. Cross-entropy in binary is preferred if the input values fall within the [0,1] range; otherwise, Mean Squared Error is employed. Variational autoencoders use a bottleneck to represent the input in a compressed format in a typical autoencoder model. Neural networks are only required as a component.

4. Results and Discussion

The experiment starts with the generation of the data since the financial institutions refused to share their data with the researcher due to strict regulations of the regulatory authority and the policies of the financial institutions to protect their customers' data, free and willing given to them. The researchers resort to synthetic data, generated with the Python programming language. The researcher uses a generative model in combination with clustering techniques to detect suspicious transactions. As part of the implementation, the generated data was loaded into Python code meant to clean the data. This will allow the dataset to fit correctly for model testing and training. Once the dataset has been loaded, several cleaning methods are used to prepare it for a particular model's training and prediction. The techniques, including identifying columns or features that will not add anything to the solution, were removed. It is followed by the deletion of single-value columns. Columns with low variance were also removed from the dataset as part of the data cleaning. Duplicate rows and columns are also removed. Outliers were identified using the following methods: outlier detection by automatic means, the interquartile range method, and the standard deviation method. Customer information, account information, and transaction details are among the input data used to identify questionable financial transactions. The transactions fall into one of two categories: suspicious or lawful. The transaction data is heavily unbalanced by nature, with suspicious transactions making up the lesser class and actual transactions making up the greater classification. The extremely unbalanced data was generated using the same approach, yielding a ratio of 95.64% to 4.36%.

4.1. Neural Network

Given a particular input, a neuron’s output is determined by mathematical formulas known as activation functions. As each node's "gatekeepers", activation functions determine how much signal should reach the following layer. The network is made non-linear by the activation functions. The ability of the neural network to gain understanding from intricate and diverse data makes this non-linearity essential. A simple linear regression model that cannot learn complex functions is distinguished from this activation function by its non-linear activation functions. Optimizing the activation function can have a big impact on how well the model works. Effective network learning can be made or broken by the activation function selection. Each activation function, such as Tanh, Sigmoid, or ReLU, has pros and cons of its own. A neuron determines whether or not to “activate” after weighing, adding a bias, and adding up all of the inputs. The “weighted sum” equation that represents this operation is commonly written as:

$$Y = sum\ (weight\_i\ * input\_i)\ + bias \tag{7}$$

How can we know if a neuron ought to activate or not, given this large range?

Activation functions are used in this exact situation. To determine the neuron’s output, activation functions act as a filter. Setting a threshold and declaring a neuron activated if Y exceeds a predetermined value may be a naïve approach to activation.

However, because this naïve method is not differentiable, we are unable to modify the weights and biases in the learning stage using backpropagation. To effectively train the network making use of gradient-based techniques like backpropagation, a plane, differentiable activation function like ReLU, Sigmoid, or Tanh is required. These functions also determine the output. A Rectified Linear Unit (ReLU), is described as follows:

$$Output = \max\ (0, x) \tag{8}$$

If the input is positive, the piecewise linear function directly produces the value; if not, it produces zero. Given that it is straight over the positive axis, ReLU may appear straight at first glance. However, ReLU is naturally not straight when taken altogether, especially as a result of the sharp corner at the start. Intriguingly, association of ReLU functions are also not straight, allowing us to store layers in neural networks efficiently.

The reason is that ReLU is a general approximator, signifying that it can mean a broad variation of functions when utilized in a neural network. A vital thing about ReLU is that it is boundless, signifying its compass is (0, ∞). While this can be functional for absolute kinds of data, it can as well make the activations to blow up if not handled well. Another notable trait of ReLU is that it appears to bring about sporadic activations. In a neural network with a lot of neurons, utilizing activation functions such as Sigmoid or Tanh would make nearly all neurons to operate to some extent, resulting in condensed activations. ReLU, alternatively, will frequently yield zero, productively disregarding some neurons, which can make the network fast in computation. However, this scarcity results in a recognized issue named the "Dying ReLU" problem. If a neuron's product is always zero (maybe as a result of a poor start), the gradient for that neuron will as well be zero.

Consequently, in backpropagation, the weights of that neuron remain unaltered, productively "eliminating" the neuron. This can bring about a part of the neural network turning unenergetic, thus reducing its ability to design complicated functions. In spite of this downside, ReLU's effectiveness and efficiency in a variety of applications continue to make it one of the most sought-after activation functions. Leaky ReLU is an altered version of the ReLU function, described as:

$$Output = for\ x\ \geq 0 \tag{9}$$

e. g. $0.015x$ for $x < 0$ for a minor constant, usually 0.01. It’s intended to solve the “Dying ReLU” issue. In the average ReLU function, the gradient turns to zero for negative inputs, leading Neurons to "die" during training. Leaky ReLU tries to fix this by establishing a small slope for negative values, typically, 0.01, to make sure the gradient is non-zero.

This little slope permits "dead" neurons to restore during training. In other words, it presents a route for gradients to move, even when the neuron is not energetic. Like traditional ReLU, Leaky ReLU is algorithmically effective. Easy mathematical operations are used, making it a little bit expensive than Tanh and Sigmoid, which is an edge in the creation of deep networks.

4.2. Loss function

The purpose of a loss function is to assess the variance in the middle of the intended and actual outputs. Take it as a mathematical tool that evaluates the disparity between what your model prognosticate and what it should normally prognosticate. This disparity is put into a single numerical value, which outlines the range of the difference. This numerical value basically evaluates the error or loss sustained by the model's prognostications.

Fundamentally, a loss function gives an indicator of how successfully your model is working against the ground truth. The lesser the loss function's value is, the nearer your model's prognostications are to the wanted results. One of the widely utilized loss functions in deep learning is the L2 norm, also referred to as the Mean Squared Error (MSE). For a given data sample, the Mean Squared Error is an indicator of the disparity betwixt the actual base truth and the output that our model predicted.

Suppose you compare a value that your model predicted to the corresponding actual value. After calculating and squaring the difference between these two, you take the average of the squared differences. The mean of the squared errors for the whole data or a subset of data is calculated, which is why it is frequently called the Mean Squared Error. This squared difference calculation is carried out for every sample in a mini-batch. A list of the individual squared errors that result can then be created.

A vector that can be used to represent this record of squared errors for every example in the mini-batch is what we refer to as $\boldsymbol{\mathcal{L}}$ or l. Mathematically, $\boldsymbol{\mathcal{L}}$ is described as a vector of each squared error, where each component of the vector denotes the squared disparity in the middle of the predicted output (x) and the ground truth (y) for a particular data point. This vector of squared errors can then be lessened to a single value by achieving the mean or the sum of the squared errors from all examples in the mini-batch.

In PyTorch, you'll frequently come across this loss function as nn.MSELoss(), that presents a fast way to calculate the Mean Squared Error in the middle of predictions and ground truths.

So, the L2 norm, via the Mean Squared Error loss function, denotes a main tool in assessing how well our model's predictions adjust with the actual data.

An alternative method of quantifying the discrepancy between the predictions of our model and the actual ground truth is the L1 norm. The L1 norm is mathematically depicted by a vector $\boldsymbol{\mathcal{L}}$, where each vector component means the absolute variation between the actual ground truth (y) and the predicted output (x) for a given data sample. The notation for this is:

$$l\_n = |x\_n - y\_n| \tag{10}$$

4.3. Generative models

[16] state that deep learning algorithms are a combination of a model, an optimization process, a cost function, and a dataset specification. After analyzing the aforementioned dataset, we will discuss the cost function, optimization, and model.

One kind of unsupervised machine learning technique for grouping data points with comparable properties is clustering algorithms. They are frequently employed in data analysis, pattern recognition, and customer segmentation. Because of its ease of use and effectiveness, the K-Means algorithm is among the most extensively utilized clustering methods. Here's a detailed explanation of how it operates:

Step 1: Decide how many clusters (k) to use.

Establishing how many clusters you wish to use to organize your data is the first step. The results of clustering are greatly impacted by the choice of k. Selecting k is not strictly regulated; instead, it frequently involves experimenting with various values and assessing the clustering's quality. In general, more specific clusters are produced by larger k values, whereas broader clusters are produced by smaller k values.

Step 2: Assign k centroids at random.

Next, the algorithm randomly chooses k data points to serve as the clusters' first centroids, which stand for the clusters' centers.

Step 3: Assign data points to the nearest centroid.

Depending on their Euclidean distance from one another, each data point is paired with the centroid that is closest to it. Consequently, k initial clusters are formed.

Continue steps 2 and 3 so that the centroids stay fixed or the cluster assignments cease shifting. This makes it possible for the clusters to converge and stabilize around the ideal centroid locations.

The sklearn.cluster API of the scikit-learn Python library makes it simple to implement K-Means and other clustering algorithms. By adjusting parameters such as the number of clusters k, various clustering configurations that best suit the underlying dataset characteristics can be investigated.

Which steps make up clustering?

Data points are grouped according to similarity using the unsupervised machine learning technique known as cluster analysis. The essential steps for implementing clustering algorithms in Python are as follows:

- Get the information libraries ready.
- Libraries such as NumPy, Pandas, Matplotlib, and Sklearn should be imported.
- Open the dataset.
- Examine the data to find any missing values, distributions, or data types.
- Data preprocessing deals with missing values, and normalize features if necessarily put together a similarity metric.
- Manhattan, Euclidean, and other distance metrics are used to quantify similarity. Select one based on the distribution and data type.
- Apply the clustering algorithm.

4.4. Results (K-Means Clustering)

Inertia (within-cluster sum of squared distances): 191.02473685317958

Silhouette Score: 0.4798814508199817

Silhouette Score: Evaluates each point's similarity to its cluster with other clusters. Values span from -1 (poor clustering) to +1 (ideal clustering).

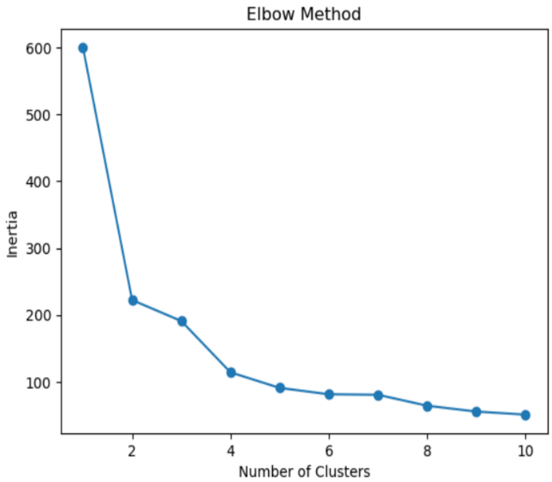

Figure 4. *The Elbow Method determines the number of clusters to use in the clustering experiment.*

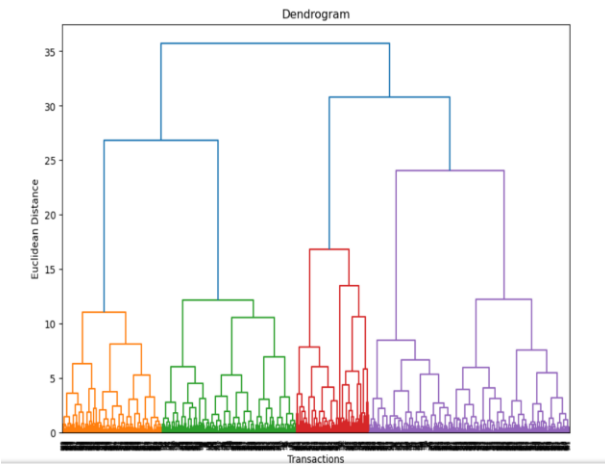

Figure 5. *Hierarchical Clustering result using the generated dataset.*

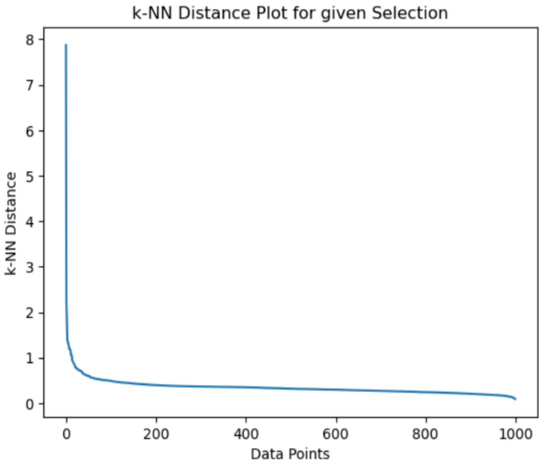

Figure 6. *DBSCAN implementation.*

Table 3. *Supervised Model's Performance*

S/N	Model	Accuracy	Precision	Recall	F1-score
1	SVM	0.980	0.960400	0.980	0.970101
2	Random Forest	0.965	0.975559	0.965	0.969638
3	XGBoost	0.970	0.976564	0.970	0.972923
4	Isolation Forest	0.940	0.9665	0.940	0.9524

Table 4. *Unsupervised Model's Performance*

S/N	Model	Silhouette	Davies-Bouldin	Calinski-Harabasz
1	K-Means	0.2681	1.4537	260.2668
2	GMM	0.2645	1.4598	259.6216
3	Hierarchical	0.2399	1.4326	246.9675

In Table 3, SVM has the best Recall (0.980). SVM captures the most positive cases, meaning it misses very few laundering events (low false negatives). It is excellent at finding suspicious transactions.

But its slightly lower precision indicates more false alarms. It is used when one wants maximum fraud detection sensitivity. Random Forest has the best Precision, 0.976. RF predictions are reliable; when it flags laundering, it's usually correct. RF has the lowest false-positive rate and slightly lower recall, and may miss some cases. It is used when avoiding false alerts is critical (compliance workload). XGBoost is the best Overall Model (F1 = 0.973). XGBoost balances precision and recall. It is better than all other models. It has strong accuracy (0.97), the Highest F1-score, the Highest precision among all supervised models (0.977), and High recall (0.970). XGBoost is used when one wants the best trade-off between catching fraud and minimizing false alarms. Isolation Forest has the weakest Supervised Performance, significantly lower accuracy (0.94), and lower recall (misses more cases). Isolation Forest is useful as an anomaly filter, but not as a strong main classifier.

K-Means is the best overall in the unsupervised Clustering Models category. K-Means has the best Overall Clustering Stability, with a decent Silhouette of 0.268, a good CH score, and a moderate Davies-Bouldin score. It is the best performing in the unsupervised method category, though it is still weak. GMM is Similar to K-Means, with almost identical performance and slightly worse on all metrics. It is good, but has no improvement over K-Means. Hierarchical clustering is the least successful of the clustering techniques that were considered. It has the lowest silhouette score (0.2399) due to the weak cluster separation [17, 18].

The performance of the models measures different parameters. By grouping data into clusters, we learn more about the internal organization of the data. Our objective in training the model may simply be to find similarity or dissimilarity in the dataset. It could be to learn about the data set underlying principles, recurring themes, and other information. Clustering is about knowledge discovery as researchers find patterns in the dataset.

Thus, clustering is subjective because similarity and dissimilarity are difficult to define. According to [19], Similarity is hard to define, but "we know it when we see it". The performance of the models indicates that XGBoost, a supervised machine learning model, is the overall best, while K-Means, in the unsupervised machine learning category, is the best in discovering patterns in the dataset.

The choice of performance measurements and evaluation criteria depends on the task, the dataset, and the specific goals of the model. For example, in a highly imbalanced classification problem, metrics like Precision, Recall, and AUC-ROC may be more important than accuracy. In contrast, if the model is intended for real-time prediction, inference time and scalability may become the most critical factors. In practice, a combination of these criterions is utilized to differentiate several machine learning algorithms and select the one that perfectly meets the needs of a given application.

4.5. Discussion

Banks used Rule-based AML systems to produce warnings in compliance with regulatory guidelines [20]. Cash transactions, foreign fund transfers, and questionable money laundering deals are all included in this. Based on the rules established by previously identified money laundering patterns, the system raises suspicious transaction alerts. Given the rate at which transactions are increasing and fraudsters are developing new fraud patterns, it is challenging to keep these regulations current [21]. This affects operational efficiency and results in more false positive alerts from the system.

Each alert should be thoroughly investigated by AML officers before being reported to the regulatory body. AML officers rely on AI/ML-based systems that generate vast amounts of data and detect suspicious transactions to assist them in making decisions. Historical money laundering patterns, such as system-generated alerts, information from case reports, and STRs, are used to train these systems. AI/ML-based AML models' primary objective is to assist AML officers by detecting skeptical deals and providing sufficient proof [20]. A bank faces severe compliance risks from unidentified suspicious transactions, which can result in severe penalties [22], according to [20]. As a result, financial institutions prioritize making sure they don't overlook any skeptical deals, despite that they need to look into more warnings. Deep learning's fundamental abilities to automatically recognize features and accurately identify suspicious transactions can reduce false positives and false negatives, which aids in the identification of suspicious transactions. For research purposes, we created synthetic data that included accounts, transactions, and customers. We saw a similar pattern because the data utilized for money laundering identification is very unbalanced, with a significant amount of legitimate transactions and a significantly smaller number of suspicious transactions. The proportions of our dataset for suspicious and legitimate transactions were 4.36% and 95.64% respectively. The high degree of imbalance in the data meant that accuracy was not regarded as a primary metric. With $\beta=3$, we valued recall over precision and looked at the $f\beta$ score as a key performance measurement metric.

5. Conclusion

Money Laundering is a global issue addressed differently across countries due to varying national laws. This study aims to identify suspicious clients using a clustering algorithm to prevent illegal funds from entering financial systems.

The research findings show that the Gaussian Mixture Model Distance algorithm outperforms Euclidean distance in detecting suspicious customer behavior. Clustering was selected for its scalability, capacity to deal with diverse data, and alignment with deep learning models to combat money laundering. The model uses minimal domain knowledge for input parameters, ensuring usability and interpretability. To prevent money laundering, financial institutions are the first line of defense. Effective AML solutions, supported by regulatory bodies such as Nigeria's NFIU, SCUML, and FATF, help banks mitigate risks. These solutions aim to automate the detection of suspicious transactions using machine learning and AI, replacing manual processes. Following the automation of many bank operations, the volume of financial transactions and client interactions has increased steadily, making AML compliance extremely difficult. A strong correspondent banking relationship between the respondent bank and the correspondent bank is necessary for the steadiness of the fiscal system and the expansion of the economies involved. International, national, and local authorities require a thorough KYC, CDD, or EDD program, which is necessary for all customers, especially politically exposed persons and their relationships [23].

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